Angular — Zero to Master Roadmap

22 topics in learning order. Follow the numbers: read the PDF in **Notes/**, type and run the programs in **Programs/**, then update the tracker.

Beginner (10 topics)

01	**Introduction to Angular**	What Angular is, why companies use it, SPAs, and the big-picture architecture.
02	**Environment Setup and Angular CLI**	Install Node.js, create a project with `ng new`, understand the folder structure and CLI commands.
03	**TypeScript Essentials**	Types, interfaces, classes, generics and modules - the language Angular is written in.
04	**Components**	Create components, use them inside other components and style them.
05	**Data Binding**	Interpolation, property, attribute, event and two-way binding between class and template.
06	**Control Flow (@if, @for, @switch)**	Show, hide and repeat parts of the template with the built-in control flow blocks.
07	**Directives, Class and Style Binding**	Change CSS classes and styles dynamically with [class], [style], NgClass and NgStyle.
08	**Signals (signal, computed, effect)**	Angular's modern reactive state: writable signals, derived values and side effects.
09	**Component Communication (input, output, model)**	Pass data down with input(), send events up with output(), and sync with model().
10	**Pipes**	Format data in templates with built-in pipes and write your own custom pipe.

Intermediate (12 topics)

11	**Services and Dependency Injection**	Share logic and data between components with injectable services.
12	**Template-Driven Forms**	Build simple forms with ngModel, ngForm and built-in HTML validation attributes.
13	**Reactive Forms**	Define forms in TypeScript with FormGroup, FormControl and FormBuilder.
14	**Form Validation and Custom Validators**	Write your own field validators and cross-field validators such as 'passwords must match'.

15	**Routing Basics**	Navigate between pages with provideRouter, router-outlet and routerLink.
16	**Route Parameters, Query Params and Child Routes**	Pass ids in URLs, read query strings and nest routes inside a layout.
17	**Route Guards and Lazy Loading**	Protect pages with functional guards and load code only when a route is visited.
18	**HttpClient and Working with APIs**	Call REST APIs with HttpClient, handle loading and error states.
19	**RxJS Fundamentals**	Observables, subscribe, pipe, map/filter and the async pipe.
20	**Component Lifecycle Hooks**	Run code at the right moment: ngOnChanges, ngOnInit, ngAfterViewInit and ngOnDestroy.
21	**Content Projection and ViewChild**	Build flexible wrapper components with ng-content and access elements with viewChild.
22	**Custom Directives**	Create your own attribute directives with inputs, outputs and host bindings.